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FSK 176

2016 Sick Test

Department of Physics, University of Pretoria
Departement Fisika, Universiteit van Pretoria

60

Date / Datum: 9/11/2016

Time / Tyd: 2.5 hours / ure

Total / Totaal: 75 max

Examiner / Eksaminator: Dr J M Nel

Moderator / Moderator: R Q Odendaal

Test register: Tear carefully on the line above and hand to the invigilator.

Toets register: Skeur versigtig op die lyn hierbo en handig dit in by die toesighouer.

FSK 176: Examination / Eksamen

20/11/2015

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Venue / Lokaal:

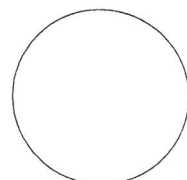
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Please read the following carefully. *Lees die volgende noukeurig*

1. Answer all the questions in the space provided on the question paper. *Antwoord alle vrae in die spasie gelaat op die vraestel.*
2. No pencil work will be marked. Do not use "tippex". *Geen potlood-werk sal gemerk word nie. Moet nie uitwis-vloeistof ("tippex") gebruik nie.*
3. If you need more space, please use the back of the pages, and clearly indicate where the question continues. *As u meer spasie benodig gebruik die agterkant van die bladsye. Dui duidelik aan waar die vraag aangaan.*
4. Where applicable you need to draw the necessary diagrams (e.g. free-body diagrams). *Teken diagramme waar nodig (bv. vryliggaams-diagramme).*
5. Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant physics. **Only substitute numerical values in the final step of calculations! A correct answer does not imply that you will receive full marks.** *U antwoorde moet sistematies en logies neergeskryf word. U moet bewys dat u die nodige fisika verstaan en kan toepas. Numeriese waardes mag slegs in die laaste stap van 'n berekening vervang word! 'n Korrekte antwoord beteken nie dat u noodwendig vol punte sal kry nie.*
6. **Units must be used when numbers are substituted else marks will be subtracted.** *Eenhede moet gebruik word wanneer numeriese waardes vervang word, andersins sal punte afgetrek word.*
7. The use of the cheat sheet is only allowed subject to the rules stated on it. *Die formuleblad mag slegs gebruik word onderhewig aan die reëls daarop gedruk.*
8. All exam regulations of the University of Pretoria apply, and any contravention of it will be treated in a serious manner. *Alle eksamen regulasies van die Universiteit van Pretoria geld en enige oortreding daarvan sal in 'n ernstige lig beskou word.*

Constants and formulas / Konstantes en formules

$\varepsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$; $G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$; $e = 1.602 \times 10^{-19} \text{ C}$; $k = 1.381 \times 10^{-23} \text{ J/K}$;
 $m_e = 9.109 \times 10^{-31} \text{ kg}$; $m_p = 1.673 \times 10^{-27} \text{ kg}$; $R = 8.314 \text{ J}/(\text{mol}\cdot\text{K})$; $N_A = 6.02 \times 10^{23} / \text{mol}$; $1 \text{ atm} = 101.3 \text{ kPa}$;
 $1 \text{ u} = 1.6605677 \times 10^{-27} \text{ kg}$; $c = 3 \times 10^8 \text{ m/s}$; $V_{\text{sphere/sfeer}} = \frac{4}{3}\pi r^3$; $F_{\text{gravity}} = \frac{Gm_1m_2}{r^2}$; $1 \text{ litre} = 1000 \text{ cm}^3$;
 $M_{\text{Earth}} = M_{\text{Aarde}} = 5.97 \times 10^{24} \text{ kg}$; $\rho(\text{water}) = 1 \text{ g/cm}^3$; $r_{\text{Earth}} = r_{\text{Aarde}} = 6.37 \times 10^6 \text{ m}$;
 $c_{\text{Al}} = 900 \text{ J/kg}\cdot^\circ\text{C}$; $\alpha_{\text{Al}} = 24 \times 10^{-6} (^\circ\text{C})^{-1}$; $L_{\text{Al},f} = 3.97 \times 10^5 \text{ J/kg}$; $\rho_{\text{Al}} = 2.70 \times 10^3 \text{ kg/m}^3$

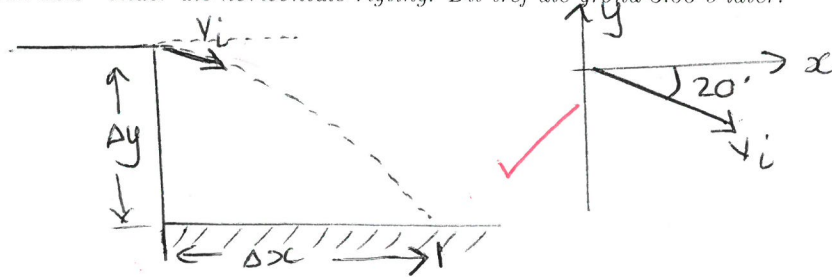


1 Question 1 / Vraag 1

[7]

A ball is tossed from the upper story window of a building with an initial speed of 8.00 m/s at an angle of 20.0° below the horizontal. It strikes the ground 3.00 s later.

'n Bal word van die boonste verdieping van 'n gebou gegooi met 'n aanvanklike spoed van 8.00 m/s teen 'n hoek van 20.0° onder die horisontale rigting. Dit tref die grond 3.00 s later.



- (a) How far horizontally from the base of the building does the ball strike the ground?

Hoe ver horisontaal vanaf die onderkant (basis) van die gebou tref die bal die grond?

(4)

$$v_{ix} = v_i \cos 20^\circ = 8.00 \text{ m/s} \cos 20^\circ$$

$$= 7.52 \text{ m/s}$$

$$\Delta x = v_{ix} t + \frac{1}{2} a_x t^2$$

$$= 7.52 \text{ m/s} (3.00 \text{ s}) + 0$$

$$= 22.6 \text{ m}$$

- (b) What is the height of the building?

Hoe hoog is die gebou?

4

$$v_{iy} = v_i \sin 20^\circ = 8.00 \text{ m/s} \sin (-20^\circ)$$

$$= -2.74 \text{ m/s}$$

$$\Delta y = v_{iy} t + \frac{1}{2} a_y t^2$$

$$= (-2.74 \text{ m/s})(3.00 \text{ s}) + \frac{1}{2} (-9.8 \text{ m/s}^2)(3.00 \text{ s})^2$$

$$= -52.3 \text{ m}$$

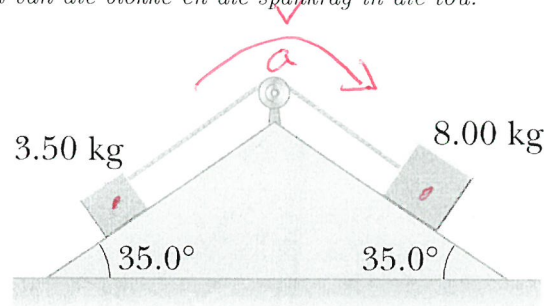
Building is 52.3 m tall ✓

2 Question 2 / Vraag 2

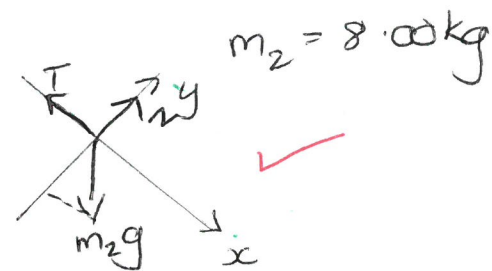
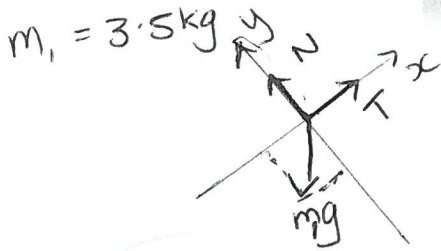
[6]

Two blocks of mass 3.50 kg and 8.00 kg are connected by a massless string that passes over a frictionless, massless pulley, as shown in the figure. The inclines are frictionless. Determine the magnitude of the acceleration of each block and the tension in the string.

Twee blokke met massas 3.50 kg en 8.00 kg word verbind deur 'n massalose tou wat oor 'n wrywinglose, massalose katrol loop, soos aangedui in die figuur. Die skuinsvlakke is wrywingloos. Bepaal die grootte van die versnelling van elk van die blokke en die spankrag in die tou.



Axes.



$$\Sigma F_{1x} = T - m_1 g \sin \theta = m_1 a$$

$$T = m_1 (a + g \sin \theta) \quad - (1)$$

$$\Sigma F_{2x} = m_2 g \sin \theta - T = m_2 a$$

$$T = m_2 (g \sin \theta - a) \quad - (2)$$

① = ② and rearrange

$$a = \frac{(m_2 - m_1) g \sin 35^\circ}{m_1 + m_2}$$

$$= \frac{(8.00 \text{ kg} - 3.50 \text{ kg})(9.8 \text{ m/s}^2) \sin 35^\circ}{(3.50 \text{ kg} + 8.00 \text{ kg})}$$

$$= 2.20 \text{ m/s}^2$$

to the right.

Then from ①

$$T = m_1 (a + g \sin \theta)$$

$$= 3.50 \text{ kg} (2.20 \text{ m/s}^2 + 9.8 \text{ m/s}^2 \sin 35^\circ)$$

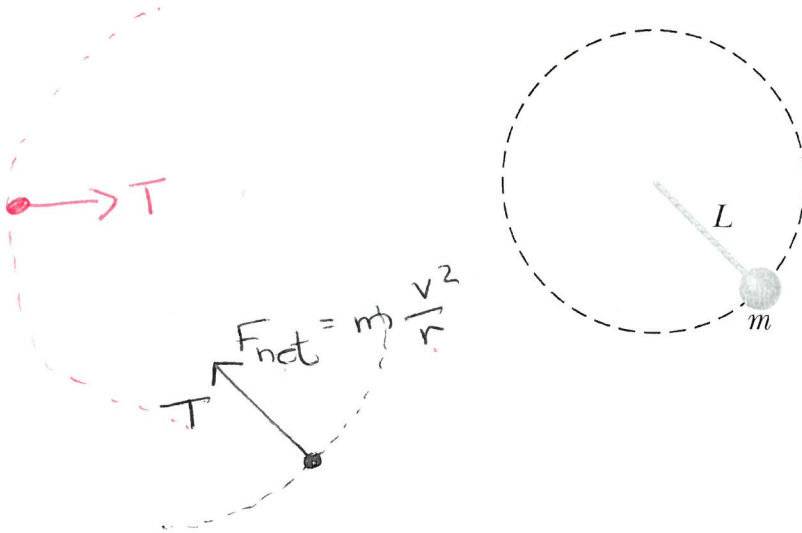
$$= 27.4 \text{ N}$$

3 Question 3 / Vraag 3

[4]

A ball of 0.500 kg is attached to the end of a cord 1.50 m long. The mass moves in a horizontal circle (parallel to the ground) as shown in the figure. If the cord can withstand a maximum tension of 50.0 N, what is the maximum speed at which the ball can move before the cord breaks? Assume the string remains horizontal during the motion.

'n Bal met massa 0.500 kg is vasgemaak aan 'n tou met 'n lengte van 1.50 m. Die massa beweeg in 'n horisontale sirkel (parallel met die grond) soos aangedui in die figuur. As die tou 'n maksimum spankrag van 50.0 N kan hanteer, wat is die maksimum spoed wat die bal kan hê voor die tou breek. Neem aan dat die tou horisontaal bly tydens die beweging.



$$m = 0.500 \text{ kg}$$

$$R = 1.50 \text{ m} = L$$

$$V = ?$$

$$T = 50.0 \text{ N}_{\text{max}}$$

$$\sum F_r = ma = m \frac{v^2}{r} = T \quad \checkmark$$

$$\therefore v = \sqrt{\frac{Tr}{m}} \quad \checkmark$$

$$= \sqrt{\frac{(50.0 \text{ N})(1.50 \text{ m})}{0.500 \text{ kg}}} \quad \checkmark$$

$$\therefore v_{\text{max}} = 12.2 \text{ m/s} \quad \checkmark$$

4

Question 4 / Vraag 4

7

A single conservative force acts on a 5.00 kg particle within a system due to its interaction with the rest of the system. The equation $F_x = 2x + 4$ describes the force, where F_x is in newtons and x is in meters. As the particle moves along the x axis from $x = 1.00$ m to $x = 5.00$ m, calculate the following:
 'n Enkele konservatiewe krag werk in op 'n 5.00 kg deeltjie as gevolg van sy interaksie met die res van die stelsel waarin die deeltjie gevind word. Die vergelyking $F_x = 2x + 4$ beskryf die krag, en F_x is in newton en x is in meter. Indien die deeltjie vanaf $x = 1.00$ m tot $x = 5.00$ m beweeg langs die x -as, bereken die volgende:

- (a) The work done by the force on the particle.
 Die arbeid verrig deur die krag op die deeltjie.

4

$$W = \vec{F} \cdot \vec{r} \quad \therefore dW = F_x dx$$

$$W = \int_{1.00 \text{ m}}^{5.00 \text{ m}} (2x + 4) dx$$

$$= \left[x^2 + 4x \right]_{1.00 \text{ m}}^{5.00 \text{ m}}$$

$$= 40.0 \text{ J}$$

- (b) The speed of the particle at $x = 5.00$ m if it has a speed of 3.00 m/s at $x = 1.00$ m.
 Die spoed van die deeltjie by $x = 5.00$ m indien die spoed 3.00 m/s is by $x = 1.00$ m.

(3)

$$W = \Delta K = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$

$$\therefore v_f^2 = \frac{2(W + \frac{1}{2} m v_i^2)}{m}$$

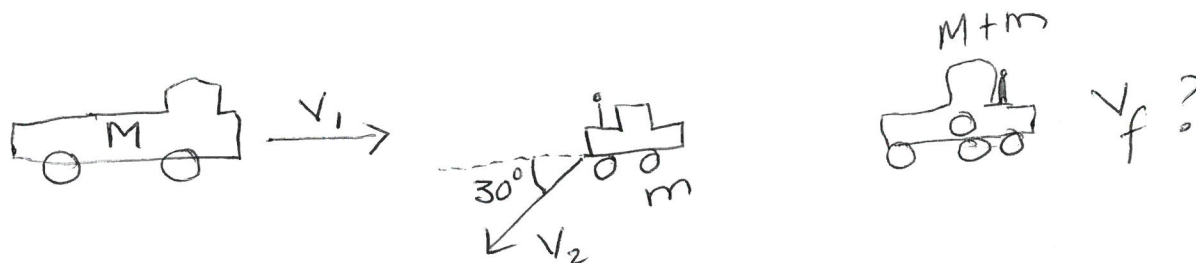
$$v_f = 5.00 \text{ m/s}$$

5 Question 5 / Vraag 5

[7]

A 7500 kg truck travelling east at 5.00 m/s collides with a 1500 kg car moving at 20.0 m/s in a direction 30.0° south of west. After the collision, the two vehicles remain tangled together. With what speed and in what direction does the wreckage begin to move after the collision?

'n 7500 kg trok beweeg ooswaarts teen 5.00 m/s en bots met 'n 1500 kg kar wat teen 20.0 m/s beweeg in 'n rigting 30.0° suid van wes. Na die botsing haak die twee voertuie aan mekaar vas. Teen watter spoed en in watter rigting begin die wrakke beweeg na die botsing?



$$P_{ix} = P_{fx}$$

$$Mv_{1x} + mv_{2x} = (M+m)v_{fx}$$

$$\frac{Mv_{1x} + mv_{2x}}{M+m} = v_{fx}$$

$$\Rightarrow v_{fx} = \frac{(7500 \text{ kg})(5.00 \text{ m/s}) - (1500 \text{ kg})(20 \cos 30 \text{ m/s})}{7500 \text{ kg} + 1500 \text{ kg}}$$

$$v_{fx} = 1.28 \text{ m/s}$$

$$P_{iy} = P_{fy}$$

$$Mv_{1y} - mv_{2y} = (M+m)v_{fy}$$

$$v_{fy} = \frac{Mv_{1y} - mv_{2y}}{M+m}$$

$$= \frac{0 - (1500 \text{ kg})(20 \sin 30^\circ \text{ m/s})}{7500 \text{ kg} + 1500 \text{ kg}}$$

$$v_{fy} = -1.67 \text{ m/s}$$

$$v_f = \sqrt{v_{fx}^2 + v_{fy}^2} = 2.10 \text{ m/s}$$

$$\theta_f = \tan^{-1} \left(\frac{v_{fy}}{v_{fx}} \right) = -52.5^\circ$$

below +x axis
south of east

6

Question 6 / Vraag 6

[6]

An electric motor rotating a grinding wheel at 1.00×10^2 rev/min is switched off. Assume the wheel has a constant negative angular acceleration of magnitude 2.00 rad/s.

'n Elektriesemotor word gebruik om 'n slypwiël te roteer teen 1.00×10^2 rev/min. Die motor word afgeskakel en jy kan aanneem dat die wiel 'n konstante negatiewe hoek-versnelling van grootte 2.00 rad/s het.

$$\omega_i = \frac{100 \text{ rev}}{\text{min}} \times \frac{2\pi \text{ rad}}{\text{rev}} \times \frac{1 \text{ min}}{60 \text{ s}} = 10.47 \text{ rad/s}$$

- (a) How long does it take the grinding wheel to stop?

Hoe lank neem dit vir die slypwiël om tot stilstand te kom?

(3)

$$\alpha = \frac{\omega_f - \omega_i}{\Delta t}$$

$$\Delta t = \frac{(\omega_f - \omega_i)}{\alpha}$$

$$= \frac{0 - 10.47 \text{ rad/s}}{-2 \text{ rad/s}^2}$$

$$= 5.24 \text{ s}$$

- (b) Through what angle has the wheel turned before coming to a stop? Give your answer in radians.

Deur watter hoek het die wiel beweeg voordat dit tot stilstand gekom het? Gee jou antwoord in radiale.

(3)

$$\omega_f^2 = \omega_i^2 + 2\alpha \Delta \theta$$

$$\Delta \theta = \frac{\omega_f^2 - \omega_i^2}{2\alpha} = \frac{0 - (10.47 \text{ rad/s})^2}{2(-2.00 \text{ rad/s}^2)}$$

$$\Delta \theta = 27.4 \text{ rad}$$

$$\boxed{\text{or}} \Delta \theta = \omega_i t + \frac{1}{2} \alpha t^2$$

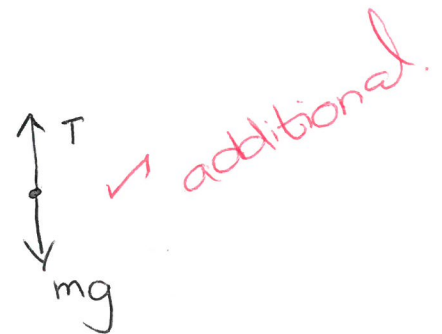
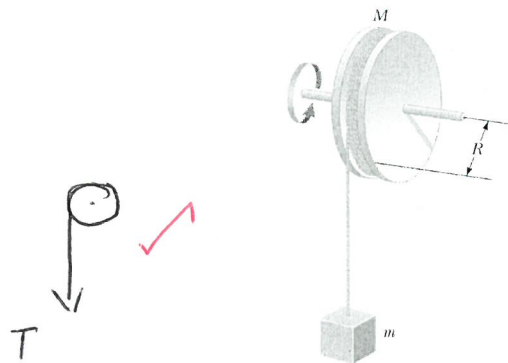
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Question 7 / Vraag 7

10

A wheel of radius R , mass M , and moment of inertia I is mounted on a frictionless horizontal axle as in the figure. A light cord wrapped around the wheel supports an object of mass m . When the wheel is released, the object accelerates downward, the cord unwraps off the wheel, and the wheel rotates with a specific angular acceleration.

'n Wiel met radius R , massa M , en traagheidsmoment I word op 'n wrywinglose, horisontale-as gemonster soos aangedui in die figuur. 'n Ligte tou word om die wiel gedraai en 'n voorwerp met massa m word van die punt van die tou gehang. Wanneer die wiel losgelaat word, versnel die voorwerp afwaarts en die tou spoel af van die wiel. Die wiel roteer met 'n sekere hoek-versnelling.



- (a) Find an expression for the angular acceleration of the wheel.

Bepaal 'n uitdrukking vir die hoek-versnelling van die wiel.

4

$$\sum \tau = I \alpha$$

$$\vec{\tau}_{\text{net}} = \vec{R} \times \vec{T}$$

$$|\vec{\tau}_{\text{net}}| = RT$$

$$\therefore RT = I \alpha$$

$$\alpha = \frac{RT}{I}$$

- (b) Find an expression for the linear (translational) acceleration of the object.
Bepaal 'n uitdrukking vir die lineêre (translasie) versnelling van die voorwerp.

(2)

$$\sum F_{\text{block}} = -ma = T - mg \quad \checkmark$$

$$a = \frac{T - mg}{-m}$$
$$= \frac{mg - T}{m} \quad \checkmark$$

- (c) Find an expression for the tension in the cord.
Bepaal 'n uitdrukking vir die spankrag in die tou.

4

$$a = \alpha R = \frac{TR^2}{I} = \frac{mg - T}{m} \quad \checkmark$$

$$\therefore \frac{TR^2 m}{I} = mg - T$$
$$T \left(1 + \frac{R^2 m}{I}\right) = mg \quad \checkmark$$

$$T = \frac{mg}{\left(1 + R^2 m / I\right)} \quad \checkmark$$

OR any other method using the
given expressions above.

Firstly, clearly state what is understood by a linear oscillator and give an example.

Then show clearly, using the basic definitions for potential and kinetic energy, that the mechanical energy E of a linear oscillator is given by $E = \frac{1}{2}kA^2$. State clearly all reasons for each step in the derivation and define all symbols used.

Eerstens, verduidelik wat 'n lineêre ossillator is en gee 'n voorbeeld daarvan.

Dan, deur gebruik te maak van die basiese definisies van potensiele energie en kinetiese energie, bewys dat die meganiese energie E van 'n lineêre ossillator gegee word deur $E = \frac{1}{2}kA^2$. Stel duidelike redes vir elke stap in jou afleiding en definieer al die simbole.

Linear oscillator: motion repeats itself in a time period T , and a x (position) eg. block on a spring.

$$U = \frac{1}{2}kx^2 = \frac{1}{2}k(A \cos(\omega t + \phi))^2$$

$$= \frac{1}{2}kA^2 \cos^2(\omega t + \phi)$$

A is amplitude.

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m \left[\frac{dx}{dt} \right]^2$$

$$= \frac{1}{2}m(-\omega A \sin(\omega t + \phi))^2$$

$$k = \omega^2 m$$

$$\therefore K = \frac{1}{2}kA^2 \sin^2(\omega t + \phi)$$

$$E_{\text{mec}} = U + K$$

$$= \frac{1}{2}kA^2 [\cos^2(\omega t + \phi) + \sin^2(\omega t + \phi)]$$

$$= \frac{1}{2}kA^2 \quad \text{Since } \sin^2 \theta + \cos^2 \theta = 1.$$

9

Question 9 / Vraag 9

[8]

A block-spring system oscillates horizontally with an amplitude of 3.50 cm. The spring constant is 250 N/m and the mass of the block is 0.500 kg.

'n Blok-veer stelsel ossilleer horisontaal met 'n amplitude van 3.50 cm. Die veerkonstante is 250 N/m en die massa van die blok is 0.500 kg.

- (a) Write an expression for the position of the oscillator as a function of time assuming that at $t = 0$ s, the oscillator is at its equilibrium position, $x = 0$ m.

Skryf die uitdrukking vir die posisie van die ossillator as 'n funksie van tyd. Neem aan dat by $t = 0$ s is die ossillator by sy ewewigspisiesie, $x = 0$ m. (3)

$$A = 3.50 \text{ cm} \quad k = 250 \text{ N/m}$$

$$\text{At } t = 0 \text{ s, } x = 0 \text{ m.}$$

$$y = A \cos(\omega t + \phi) \quad \phi = \pi/2$$

$$\omega = \sqrt{k/m} = \sqrt{250 \text{ N/m} / 0.500 \text{ kg}} \\ = 22.4 \text{ rad/s}$$

$$\therefore y = 3.50 \text{ cm} \cos(22.4 t + \pi/2) \\ = 3.50 \text{ cm} \sin(22.4 t)$$

- (b) Determine the mechanical energy of the system.

Bepaal die meganiese energie van die stelsel. (2)

$$E = \frac{1}{2} k A^2 = \frac{1}{2} (250 \text{ N/m}) (3.50 \times 10^{-2} \text{ m})^2 \\ = 0.153 \text{ J}$$

- (c) Determine the maximum acceleration of the block and at what position this occurs.
Bepaal die maksimum versnelling van die blok en by watter posisie dit plaasvind.

(3)

This occurs at the maximum displacement, A or $-A$.

$$a_{\max} = \omega^2 A$$

$$= (22.4 \text{ rad/s})^2 (3.5 \times 10^{-2} \text{ m})$$

$$= 17.6 \text{ m/s}^2$$

End of sick Test

10

Question 10 / Vraag 10

$kx + \omega t$

[8]

A transverse wave on a string is described by the wave function $y = 0.120 \sin(\frac{\pi}{8}x + 4\pi t)$ where x and y are in meters and t is in seconds. Determine the following:

'n Transversale golf in 'n tou word beskryf deur die golf-funksie $y = 0.120 \sin(\frac{\pi}{8}x + 4\pi t)$ waar x en y in meter is en t in sekondes is. Bepaal die volgende:

- (a) The transverse acceleration at $t = 2 \text{ s}$ for an element of the string located at $x = 1.60 \text{ m}$.

Die transversale versnelling by $t = 2 \text{ s}$ vir 'n element van die tou geleë by $x = 1.60 \text{ m}$.

(3)

$$\omega = 4\pi \quad k = \pi/8$$

$$y = 0.12(4\pi) \cos(\pi/8 x + 4\pi t)$$

$$a = -0.12(4\pi)^2 \sin(\pi/8 x + 4\pi t)$$

At $t = 2 \text{ s}$ and $x = 1.6 \text{ m}$

$$a = - (0.12)(4\pi)^2 \sin(\frac{\pi}{8}(1.60 \text{ m}) + 4\pi(2)) \text{ m/s}^2$$

$$= -11.14 \text{ m/s}^2$$

- (b) The wavelength of the wave.
Die golf-lengte van die golf.

(2)

$$k = \frac{2\pi}{\lambda} \Rightarrow \lambda = \frac{2\pi}{k}$$

$$\lambda = \frac{2\pi}{\pi/8} = 16.0 \text{ m}$$

- (c) If the string is clamped on both ends and the wave described above is now reflected and a standing wave forms, what will the resulting **amplitude** be and what must the **length** of the string be to allow 5 nodes to be formed?

As die tou aan beide kante vasgemaak word, en die bostaande golf word nou terug-gereflekteer en 'n staande-golf vorm, wat sal dié resultante **amplitude** wees en wat moet die tou se **lengte** wees om 5 nodes te vorm op die tou?

(3)

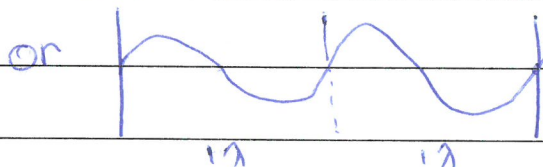
Standing wave amplitude is $2A$.

$$\therefore \text{Amplitude} = 2(0.120) \text{ m} \\ = 0.240 \text{ m}$$

For 5 nodes, 4 antinodes $\Rightarrow n = 4$

$$\lambda = \frac{2L}{n}$$

$$L = \frac{n\lambda}{2} = \frac{(4)(16.0 \text{ m})}{2} = 32 \text{ m}$$

or  5 nodes = $2\lambda = 32 \text{ m}$.

11 Question 11 / Vraag 11

[10]

A 1.00 kg rod of aluminium is warmed up at atmospheric pressure so that its temperature increases from 22.0°C to 40.0°C . Determine:

'n 1.00 kg staaf aluminium word by atmosferiese druk warm gemaak sodat sy temperatuur van 22.0°C tot 40.0°C styg. Bepaal:

- (a) The work done on the rod.
Die arbeid verrig op die staaf.

(4)

Constant Pressure $\therefore W = -P \Delta V$

$$\Delta V = V\beta \Delta T = \left(\frac{m}{\rho}\right)(3\alpha)(\Delta T) =$$

$$= \left(\frac{1.00 \text{ kg}}{2.70 \times 10^3 \text{ kg/m}^3}\right)(3 \times 24 \times 10^{-6} / ^\circ\text{C})(40^\circ\text{C} - 22^\circ\text{C})$$

$$\Delta V = 4.8 \times 10^{-7} \text{ m}^3 \quad \checkmark$$

$$W = -P \Delta V = -101.3 \times 10^3 \text{ Pa} \times 4.8 \times 10^{-7} \text{ m}^3 \\ = -4.86 \times 10^{-2} \text{ J} \quad \checkmark$$

- (b) The energy added to the rod in the form of heat.

Die energie wat in die vorm van warmte oorgedra is aan die staaf.

(3)

$$Q = m c \Delta T \\ = (1.00 \text{ kg}) (900 \text{ J/kg} \cdot ^\circ\text{C}) (40^\circ\text{C} - 22^\circ\text{C}) \quad \checkmark \checkmark \\ = 16.2 \times 10^3 \text{ J} \\ = 16.2 \text{ kJ} \quad \checkmark$$

- (c) The change in the internal energy of the aluminium rod.

Die verandering in die interne energie van die aluminium staaf.

(3)

$$\Delta E_{\text{int}} = Q + W \quad \checkmark$$

$$= (16.2 \times 10^3 \text{ J}) + (-4.86 \times 10^{-2} \text{ J}) \quad \checkmark$$

$$= 16.2 \times 10^3 \text{ J} = 16.2 \text{ kJ.} \quad \checkmark$$

THE END OF FSK176 EXAM / DIE EINDE VAN FSK176
EKSAMEN